Pietro Mesquita Piccione

Résumé

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Work experience

2025 – 2027 **Postdoc**, *University of Gothenburg*, Göteborg, Sweden, Supervisor: David Witt Nyström

Education

2022 – 2025 ***PhD in Mathematics***, *Institut des mathématiques de Jussieu*, Sorbonne Université, Paris, France
Advisors: Sébastien Boucksom and Tat Dat Tô

2021 – 2022 ***Master's Degree in Mathematics***, *Institut des mathématiques de Jussieu*, Sorbonne Université, Paris, France

2017 – 2021 ***Bachelor's degree in Mathematics***, *Universidade de São Paulo*, USP, São Paulo, Brazil
Prize for outstanding performance.

Research projects and grants

2026 – 2027 **Research grant: "Kähler and non-Archimedean geometry"**, *Funded by Stiftelsen Wilhelm och Martina Lundgrens Vetenskapsfond*
Grant of 50.000 SEK for research purposes.

2022 – 2025 **MathInParis2020**, *Scholarship for a PhD in Paris*, Cofunded by Marie Sklodowska-Curie Actions, Fondation sciences mathématiques de Paris

2021 – 2022 **PGSM Scholarship**, *Scholarship for a Master's degree in Paris*, Fondation sciences mathématiques de Paris

2020 – 2021 **Undergraduate research project: "Morse Theory"**, *Funded by the São Paulo Research Foundation, FAPESP*
Grant FAPESP 2020/04871-0

Articles

2026 , *joint with David Witt Nyström*, In preparation
description

2025 **A transcendental non-Archimedean Calabi–Yau Theorem with applications to the cscK problem**, *joint with David Witt Nyström*, preprint, arXiv:2509.09442
Description: We solve the non-Archimedean Monge-Ampère equation for compact Kähler manifolds, generalizing a result of Boucksom–Favre–Jonsson to the transcendental setting. This allows us to derive a valuative criterion for uniform K-stability for models. As a central application, we prove that this stability condition implies the existence of a unique constant scalar curvature Kähler metric.

2024 **A non-Archimedean theory of complex spaces and the csc-K problem**, *published*, Advances in Mathematics
Description: In this paper I construct a space of valuations attached to a compact complex manifold, that generalize the Berkovich analytification of a projective variety over $\mathbb{C}$. As an application I get a sufficient algebraic condition for the existence of constant scalar curvature Kähler metrics on a given Kähler class on a Kähler manifold.

Professional

Supervision

2026 **Undergraduate research project: the geometry of Maxwell's equations**, *Main supervisor of Andy Montalvo, under the Irere project*, Online

Teaching

2026 **Riemann Surfaces**, *PhD course*, U. of Gothenburg and Chalmers University of Technology, Sweden

Teaching Assistantships

2025 **Introduction to Riemann Surfaces**, *Master's course*, Sorbonne Université, France

2024 **Topology and Differentiable Calculus I**, *Second year bachelor's course*, Sorbonne Université, France

2024 **Functional Analysis**, *Third year bachelor's course*, Sorbonne Université, France

2023 **Topology and Differentiable Calculus I**, *Second year bachelor's course*, Sorbonne Université, France

2020 **Introduction to Algebraic Topology**, *Fourth year bacherlor's course*, Universidade de São Paulo, Brazil

2019 **Introduction to Analysis**, *Third year teacher's diploma course*, Universidade de São Paulo, Brazil

2019 **Differentiable functions and series**, *Second year Computer Science's course*, Universidade de São Paulo, Brazil

2018 **Calculus II**, *First year Engineering course*, Universidade de São Paulo, Brazil

Events Organized

2026 **Winter Geometry Workshop**, *3 days workshop on complex geometry*, University of Gothenburg, Sweden

2026 **Geometry and Topology of (Almost) Complex Structures**, *Online conference*

2022 **MaThematical Days : Paul Erdős**, *A day workshop on the life and work of Paul Erdős*, Universidade de São Paulo, Brazil

2021 **MaThematical Days : Emmy Noether**, *A day workshop on the life and work of Emmy Noether*, Universidade de São Paulo, Brazil

2021 **MaThematical Days : Élie Cartan**, *A day workshop on the life and work of Élie Cartan*, Universidade de São Paulo, Brazil

Scientific Visits

2025 **Simons Semester in Analysis and Geometry on Complex Manifolds**, *Erdős Center*, Budapest, Hungary
Three months visit to the program.

2024 **Special Geometric Structures and Analysis**, *SLMath*, Berkeley
Two months visit to the program.

Academic Services

2025 **Participated at the cycle "Un texte, une aventure mathématique"**, *Lycée international St Germain en Laye*, Slides and material
Description: Gave a "pre-talk" at a high school preparing for the talk "Dennis Sullivan, mathématicien des analogies" given by Barbara Schapira. "Un texte, une aventure mathématique" is a joint organization of the Société Mathématique de France and the Bibliothèque nationale de France to expose the general public with the history and problems of current research Mathematics.

2021 **Organizer of the A5 Group**, Instituto de Matemática e Estatística, Universidade de São Paulo
A student runned academic group with over 25 weekly seminars organized

2018 – 2019 **Deputy Student Representative in the Mathematics Department Council**, Instituto de Matemática e Estatística, Universidade de São Paulo, Brazil

Talks

A non-Archimedean Calabi–Yau theorem for Kähler manifolds.

2026 **Degenerations of complex manifolds**, *Institute Mittag Leffler*, Stockholm, Sweden

2026 **Seminario di Geometria**, *Università degli studi di Roma Tor Vergata*, Rome, Italy

$\widehat{K}$-stability on Kähler manifolds

2026 **Workshop on K-stability**, East China Normal University, Shanghai, China

The non-Archimedean approach of the Yau–Tian–Donaldson conjecture

2026 **Geometry, HOlonomy and Structures with Torsion**, University of Münster, Münster, Germany

2026 **Geometric analysis seminar**, Westlake University, Hangzhou, China

2026 **Geometry and topology seminar series**, University of Science and Technology of China, Hefei, China

2025 **Séminaire de géométrie**, Centre de mathématiques Laurent-Schwartz, Palaiseau, France

2025 **Workshop on Singular canonical Kähler metrics on compact and non-compact manifolds**, Rényi Institute, Budapest, Hungary

2025 **Geometry seminar**, Max Plank Institute for Mathematics in the Sciences, Leipzig, Germany

2025 **Complex geometry seminar**, Mathematisches Institut, University of Münster, Munster, Germany

2025 **MARGE in Brest: Fibrations and Deformations**, Université de Bretagne Occidentale, Brest, France

2025 **Geometry Webinar AmSur /AmSul**, Webinar

2025 **Séminaire de Géométrie complexe**, Université Toulouse III - Paul Sabatier, Toulouse, France

2025 **Séminaire Géométries ICJ**, Université Claude Bernard Lyon 1, Lyon, France

2025 **Mathematics Seminar**, Aarhus University, Aarhus, Denmark

2024 **Séminaire de géométrie algébrique de Jussieu**, ENS, Paris, France

2024 **Séminaire systèmes dynamiques et géométrie**, Université d'Angers, Angers, France

2024 **Complex Analysis Seminar**, University of Bayreuth, Bayreuth, Germany

2024 **Geometric Analysis Seminar**, BICMR, Beijing, China

2024 **KASS Seminar**, Chalmers University, Gothenburg, Sweden

2024 **Algebraic Geometry Seminar**, University of Glasgow, Scotland

2024 **Any Complex Geometry Seminar**, University of California Berkeley, US

2024 **Graduate Student Seminar Series**, SLMath, Berkeley, US

A counterexample for the periodic orbit conjeture

2022 **Groupe de Travail : Dynamiques Sauvages**, IMJ-PRG, Sorbonne Universite, France

2022 **A5 seminar**, Instituto de Matemática e Estatística, Universidade de São Paulo, Brasil

To the general public

2021 **A Tale of Geometry**, *S4*, *Instituto de Matemática e Estatística*, Universidade de São Paulo, Brazil

2020 **Morse Theory in 15 Minutes**, *Senior Thesis Presentation*, *Instituto de Matemática e Estatística*, Universidade de São Paulo, Brazil

2020 **Is the Earth flat? A talk about curvature**, *S4*, *Instituto de Matemática e Estatística*, Universidade de São Paulo, Brazil

2020 **How to solve any equation, with some conditions**, *S4*, *Instituto de Matemática e Estatística*, Universidade de São Paulo, Brazil

Other Info

Related Activities

2025 **Reading Seminar on Kähler Ricci Flow**, Online

2021 – 2022 **Reading Seminar on Wild Dynamics**, *Groupe De Travail Dynamiques Sauvages*, IMJ-PRG, Sorbonne Université

Languages

Portuguese	Native Speaker
Italian	Native Speaker
English	Fluent
Spanish	Fluent
French	Fluent
Catalan	Intermediate

Memberships

2026 Member at the Société Mathématique de France

Links

ArXiv https://arxiv.org/a/mesquitapiccione_p_1

Lattes http://lattes.cnpq.br/7822745238118143
Orcid https://orcid.org/0000-0002-1626-0726